SOLUTION

SECTION–I

1. Ans. (C)

As the point $(3\sqrt{3}, 2)$ lies on the line $y = \sqrt{3}x - 7$ so locus of P will be a line perpendicular to given line and passing through $(3\sqrt{3}, 2)$

$$y - 2 = -\frac{1}{\sqrt{3}}(x - 3\sqrt{3})$$

$$y - 2 = \frac{-x}{\sqrt{3}} + 3$$

$$\frac{x}{\sqrt{3}} + y - 5 = 0$$

It intersects x and y axis at $(5\sqrt{3}, 0)$ and $(0, 5)$

so area of the triangle formed $= \frac{5\sqrt{3} \times 5}{2} = \frac{25\sqrt{3}}{2}$

2. Ans. (B)

$$\frac{y_2 - y_1}{x_2 - x_1} + \frac{y_3 - y_2}{x_3 - x_2} + ... + \frac{y_n - y_{n-1}}{x_n - x_{n-1}}$$

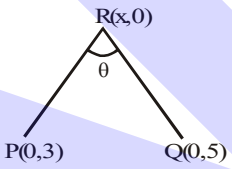
because $x_2 - x_1 = x_3 - x_2 = ... x_n - x_{n-1} = d$

$$\frac{y_n - y_1}{d} = \frac{(n-1)(y_n - y_1)}{(n-1)d} = \frac{(n-1)(y_n - y_1)}{(x_n - x_1)} = (n-1)m_{A_1A_n}$$

3. Ans. (A)

$$m_1 = -\frac{3}{x}$$

$$m_2 = -\frac{5}{x}$$



$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{\frac{2}{x}}{1 + \frac{15}{x^2}}\right|$$

$$\tan\theta = 2\left|\frac{x}{15 + x^2}\right|$$

$$\lambda = \frac{x}{15 + x^2}$$

$$\frac{d\lambda}{dx} = \frac{(15 + x^2) - 2x^2}{(15 + x^2)^2} = 0$$

$$x^2 = 15 \Rightarrow |x| = \sqrt{15}$$

4. Ans. (B)

Let the point of intersection of tangents at A & B is (x_1, y_1), then equation of chord of contact of (x_1, y_1) w.r.t. given circle is $T = 0$

$\Rightarrow$ $xx_1 + yy_1 - 3(x + x_1) - 2(y + y_1) + 3 = 0$

$\Rightarrow$ $x(x_1 - 3) + y(y_1 - 2) - 3x_1 - 2y_1 + 3 = 0$

and equation of chord whose mid point is (1, 1) is

$T = S_1 \Rightarrow$ $x + y - 3(x + 1) - 2(y + 1) + 3 = 1 + 1 - 6 - 4 + 3$

$\Rightarrow$ $-2x - y + 3 = 0$

As both equations are representing the same straight line

$$\Rightarrow \frac{x_1 - 3}{-2} = \frac{y_1 - 2}{-1} = \frac{-3x_1 - 2y_1 + 3}{3}$$

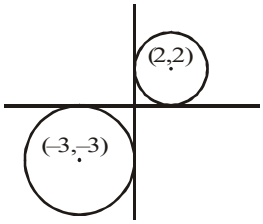
$\Rightarrow$ $x_1 = -1$, $y_1 = 0$

5. Ans. C

$x^2 + y^2 - 4x - 4y + 4 = 0$ $c = (2, 2)$ $r = 2$

$x^2 + y^2 + 6x + 6y + 9 = 0$ $c = (-3, -3)$ $r = 3$

From figure it is clear that



tangents are $x = 0$; $y = 0$

6. Ans. (B)

Bisectors of the transverse common tangents of $(x \pm 2)^2 + y^2 = 1$ are x & y–axis

$\therefore$ Let the centre of the required circle be (r, r).

$\therefore$ It touches $(x - 2)^2 + y^2 = 1$

$\therefore$ $c_1c_2 = r_1 + r_2$

$\Rightarrow (r - 2)^2 + r^2 = (r + 1)^2 \Rightarrow r^2 - 6r + 3 = 0$

$$\Rightarrow r = \frac{6 \pm 2\sqrt{6}}{2}$$

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$r = 3 \pm \sqrt{6}$

7. Ans. (D)

(a,3) lies inside $x^2 + y^2 - 16 = 0$ and outside $x^2 + y^2 - 4 = 0$ and above y = x in first quadrant so

$a^2 + 9 - 16 < 0 \Rightarrow -\sqrt{7} < a < \sqrt{7}$ ...(i)

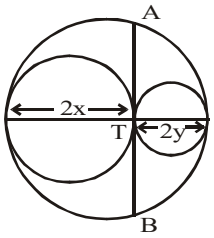
$a^2 + 9 - 4 > 0 \Rightarrow a \in R$ ...(ii)

$a - 3 < 0 \Rightarrow a < 3$ ...(iii)

in first quadrant $a > 0$ ...(iv)

from (i), (ii), (iii) & (iv) $\Rightarrow a \in (0, \sqrt{7})$

8. Ans. (D)



Let T is point of contact

power of point

$= TA \cdot TB = 2.2 = 2x \cdot 2y$

$\Rightarrow \quad xy = 1$

we have to find

$\pi\left[(x+y)^2 - x^2 - y^2\right] = \pi[2xy] = 2\pi$

9. Ans. (A)

$-f = -g - 2 \qquad -g = -2f$

$f = g + 2$ ..... (i) $\qquad g = 2f$ ..... (ii)

$f = 2f + 2$

$\therefore \quad f = -2 \quad \& \quad g = -4$

$g^2 - c = 0 \Rightarrow \quad c = 16$

$\therefore$ Equation of the required circle is

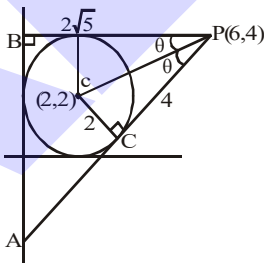
$x^2 + y^2 - 8x - 4y + 16 = 0$

10. Ans. (B)

Circle is $(x-2)^2 + (y-2)^2 = 2^2$

BP = 6

$\Rightarrow$ CP = 4



$\tan\theta = \frac{1}{2}$

$\tan 2\theta = \frac{AB}{BP}$

$\Rightarrow \quad AB = 6\tan 2\theta$

$= \frac{12\tan\theta}{1-\tan^2\theta} = \frac{6}{1-\frac{1}{4}} = 8$

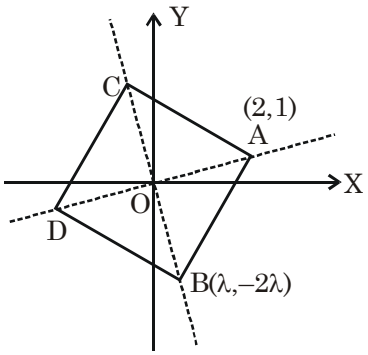
11. Ans. C,D

Sol. $2x^2 - 3xy - 2y^2 = 0$

$2x^2 - 4xy + xy - 2y^2 = 0$

$(2x + y)(x - 2y) = 0$

$\therefore$ diagonals are $y = \frac{x}{2}$ and $y = -2x$



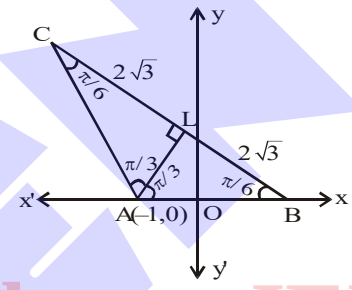
A(2, 1) and B(λ, –2λ)

$OA^2 = OB^2$

$\lambda = \pm 1$

$\therefore$ B(1, –2) or C(–1, 2)

12. Ans. (A,B,D)



In ΔABC, we have

$BL = AB\cos\frac{\pi}{6}$

$2\sqrt{3} = AB \times \frac{\sqrt{3}}{2} \quad \Rightarrow \quad AB = 4 \quad \Rightarrow \quad OB = 3$

So, the co-ordinates of B are (3, 0)

So the equation of BC is

$y - 0 = \tan\frac{5\pi}{6}(x - 3)$

$\Rightarrow \quad y = -\frac{1}{\sqrt{3}}(x-3) \quad \Rightarrow \quad x + \sqrt{3}y - 3 = 0$

$\Delta = \frac{1}{2}AB.AC.\sin\frac{2\pi}{3} = 4\sqrt{3}$

13. Ans. (B,C,D)

$S_1 : x^2 + y^2 - 4x - 6y - 12 = 0$

$r_1 = \sqrt{4 + 9 + 12} = 5$

$S_2 : x^2 + y^2 + 6x + 4y - 12 = 0$

$r_2 = \sqrt{9 + 4 + 12} = 5$

$S_1 - S_2 = x + y = 0$

$C_1C_2 = \sqrt{25 + 25} = 5\sqrt{2}$ Hence

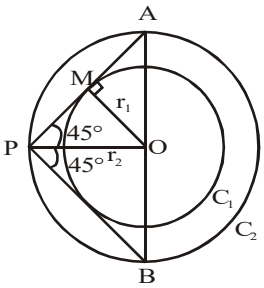
(B) (C) (D)

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14. Ans. (B,D)

In ΔOPM, we have



$$\sin 45^\circ = \frac{OM}{OP}$$

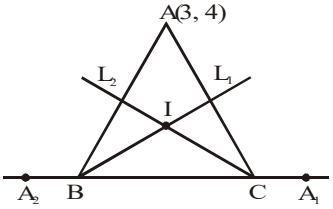
$$\Rightarrow \quad \frac{1}{\sqrt{2}} = \frac{r_1}{r_2}$$

$$\Rightarrow \quad r_2 = \sqrt{2} r_1$$

Paragraph for Question 15 to 16

15. Ans. (B)

Let the image of point A with respect to L_2 is A_2



let A_1 is (α, β)

Now equation of the line perpendicular to L_1 and passing through (3, 4) will be

$y - 4 = x - 3 \quad \Rightarrow \quad x - y + 1 = 0$

Intersection point of this line & L_1 is $\left(\frac{1}{2}, \frac{3}{2}\right)$

Now $\frac{\alpha + 3}{2} = \frac{1}{2}$ & $\frac{\beta + 4}{2} = \frac{3}{2}$

$\alpha = -2, \qquad \beta = -1$

$\therefore \quad A_1 \equiv (-2, -1)$

Similarly A_2 will be $(-1, 2)$

from geometry A_1, A_2 will lie on BC

so slope of BC will be 3.

16. Ans. (C)

Intersection point of L_1 & L_2 will be incentre

i.e. $(3, -1)$

equation of BC will be

$y - 2 = 3(x + 1)$

$3x - y + 5 = 0$

Now the inradius will be

$$\frac{9 + 1 + 5}{\sqrt{10}} = \frac{15}{\sqrt{10}} = \frac{3}{2}\sqrt{10}$$

Paragraph for Question 17 to 18

$C_1 (0,0)$; $C_2(6,0)$

$r_1 = 1 \qquad r_2 = 2$

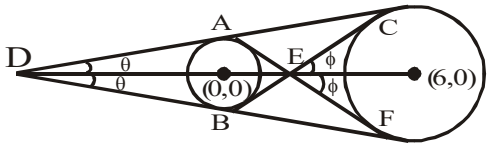
Distance between centres $d = 6$

17. Ans.(C)

Length of direct common tangent

$$= \sqrt{d^2 - (r_2 - r_1)^2} \quad = \sqrt{36 - 1} = \sqrt{35}$$

18. Ans.(B)



Both centres lie on x-axis

$\therefore$ Intersection point of direct common tangent & Intersection point of transverse common tangent lie on x-axis

Slope of tangent DA = $\tan\theta$

Slope of tangent DB = $-\tan\theta$

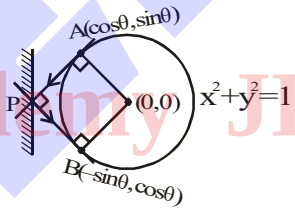
Slope of tangent CE = $\tan\phi$

Slope of tangent FE = $-\tan\phi$

$\therefore$ Sum of slope of all common tangents is 0.

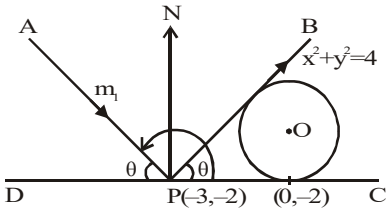
SECTION–III

1. Ans. 2



It is obvious that tangents are perpendicular, then locus of P is its director circle $x^2 + y^2 = 2$.

2. Ans. 7



Any line through P(–3,–2) is

$y + 2 = m(x + 3)$

It touches the circle if $mx - y + 3m - 2 = 0$

$$\left|\frac{3m - 2}{\sqrt{1 + m^2}}\right| = 2$$

$9m^2 + 4 - 12 = 4 + 4m^2$

$5m^2 - 12m = 0$

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$m = 0, \frac{12}{5}$

Equation of PB is

$y + 2 = \frac{12}{5}(x + 3)$

$5y + 10 = 12x + 36$

$12x - 5y + 26 = 0$

Slope $m = \frac{12}{5}$

$$\tan\theta = \left|\frac{\frac{12}{5} - 0}{1 + 0}\right|$$

$\tan\theta = \frac{12}{5}$

$\angle CPB = \angle DPA$

$\therefore \angle CPA = \pi - \theta$

$\tan(\pi - \theta) = -\tan\theta$

$\tan(\pi - \theta) = -\frac{12}{5}$

Let slope of AP is m_1

$m_1 = -\frac{12}{5}$

So equation of AP is

$y + 2 = -\frac{12}{5}(x + 3)$

$5y + 10 = -12x - 36$

$12x + 5y + 46 = 0$

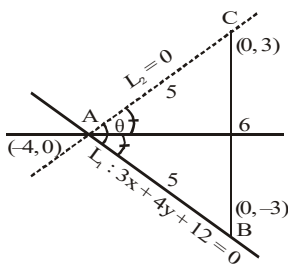
$Jx - ky + 46 = 0$

$J = 12 \quad K = -5$

$J + K = 7$

3. Ans. 9

$\theta = \tan^{-1}\left(\frac{24}{7}\right)$



$\frac{24}{7} = \left(\frac{m - (-3/4)}{1 + m(-3/4)}\right)$

$m = \frac{3}{4}$

$\therefore$ equation of the line $L_2 = 0$ is

$L_2 : 3x - 4y + 12 = 0$

Now, incentre of the ΔABC is

$$\left(\frac{5\times0 + 6\times(-4) + 5\times0}{16}, \frac{5\times3 + 6\times0 + 5\times(-3)}{16}\right)$$

$\equiv \left(-\frac{3}{2}, 0\right) \equiv (4\lambda, 3k)$

$\therefore \quad \lambda = -\frac{3}{8}, \quad k = 0$

$10k - 24\lambda = 9$

4. Ans. 2

$d(P, OB) \le d(P, AB)$

$d(P, OB) \le d(P, OA)$

P must lie in the triangle OIB where I is incentre of the triangle OAB

$$I \equiv \left(\frac{3\times3 + 3\times5 + 4\times0}{3+4+5}, \frac{4\times3 + 0\times5 + 0\times4}{3+4+5}\right)$$

$\equiv (2, 1)$

Number of integral points inside the triangle OIB is 2, which are (1, 1) & (2, 2)

SECTION–IV

1. Ans. (A)→(S), (B)→(R), (C)→(P), (D)→(Q)

(A) Equation of common chord

$10x + 4y - b - a = 0$

this chord should pass through the centre of the circle $x^2 + y^2 + 6x + 6y - b = 0$

i.e. $(-3, -3)$

$-30 - 12 - b - a = 0$

$a + b = -42$

$|a + b| = 42$

(B) Equation of chord of contact drawn from point (a, b) will be $ax + by = 10$

comparing it with given chord

$x + y = 2$

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$a = 5,\ b = 5$

so $a^2 + b^2 = 50$

(C) Equation of the family of circles passing through AB will be

$x^2 + y^2 - 5x + \lambda(2x - y) = 0$

$\Rightarrow x^2 + y^2 + x(2\lambda - 5) - \lambda y = 0$

centre $\left(-\frac{(2\lambda - 5)}{2}, \frac{\lambda}{2}\right)$

If AB is a diameter then centre must lie on AB, so

$\frac{\lambda}{2} = -(2\lambda - 5)$

$\lambda = -4\lambda + 10$

$\lambda = 2$

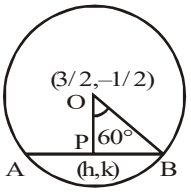
$\therefore$ required circle will be

$x^2 + y^2 - x - 2y = 0$

$a = 1,\ b = 2$

$\therefore\ 2a + 9b = 20$

(D) Let mid point of the chord is (h, k)



In ΔOPB

$\cos 60^\circ = \frac{OP}{OB}$

$\frac{1}{2} = \frac{\sqrt{(h - 3/2)^2 + (k + 1/2)^2}}{\sqrt{2}}$

$\Rightarrow (h - 3/2)^2 + (k + 1/2)^2 = 1/2$

$\Rightarrow h^2 + k^2 - 3h + k + 2 = 0$

$\therefore$ Required locus $x^2 + y^2 - 3x + y + 2 = 0 \Rightarrow a = 3,\ b = -1$

So $a^3 + b^3 = 27 - 1 = 26.$

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